Week 05: Machine Learning

Data Science Bootcamp
Summer, 2021

Instructor: Sagar Patel

Almost done...

Week 1

Introduction

Introduction to Python; Getting started with Git and GitHub

Week 2

SQL

Relational databases; Selecting, joining and updating tables using SQL

Week 3

Python libraries

Python packages and libraries; Working with numpy, pandas, matplotlib and seaborn

Week 4

Exploratory Data Analysis

What makes data good? The “recipe” for performing EDA

Week 5

Machine Learning

Week 6

Project & Wrap

Agenda

- Machine Learning
    - What is it?
    - How vast does ML get?
    - Linear and Logistic Regression
    - Clustering
    - Model representation and Cost Function

NOTE: Machine Learning is a huge topic! We will barely scratch the surface in this session

Machine Learning

slido

What are some real-world examples where Machine Learning is used?

ⓘ Start presenting to display the poll results on this slide.

A few examples

Recommendation System

Image Classification

NLP

slido

Suppose your email program watches you marking emails as spam/not spam, and based on this it learns to make the filter spam better.

What is the task in this setting?

ⓘ Start presenting to display the poll results on this slide.

What is Machine Learning?

- The ability to **learn** from **data**, without being explicitly programmed using **statistical techniques**

What is Machine Learning?

- The ability to **learn** from **data**, without being explicitly programmed using **statistical techniques**

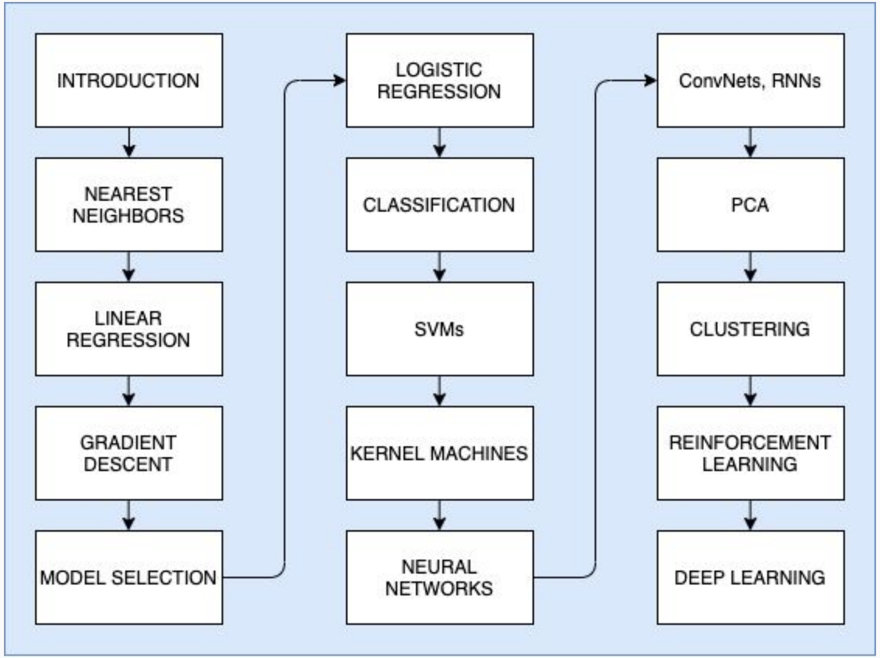
- Alternatively,
  `f(data) = Actionable Information`

Ingredients of Machine Learning

Follow the 3-step recipe while solving any Machine Learning problem:

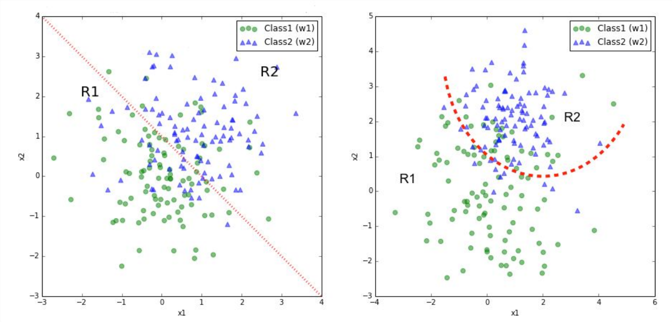
- Data **Representation**
- Measure of “**goodness**”
  - Loss function
- A **method** for optimizing the measure of goodness
  - **Training** methods
  - Manipulation using features of the data

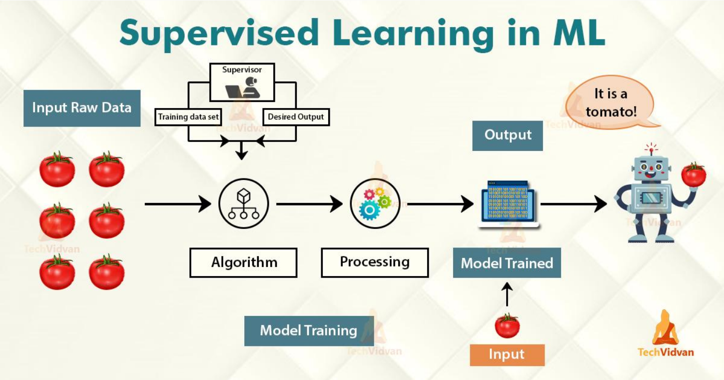
How “vast” is Machine Learning?



Types of “Learning”

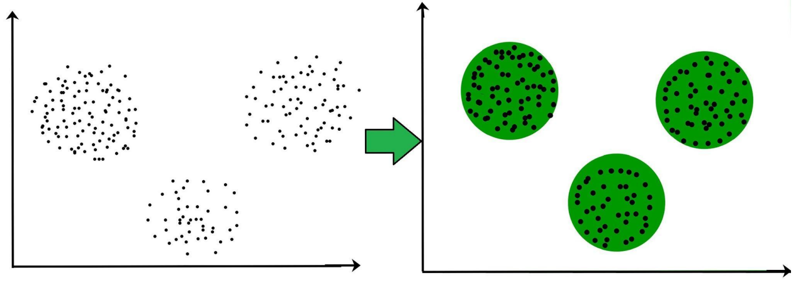
- **Supervised Learning**
    - Algorithms trained using **labeled** data
    - The model takes direct **feedback** to check if it’s predicting the correct output or not
    - Can be categorized in **Classification** and **Regression**
    - Example: Tomato Detector

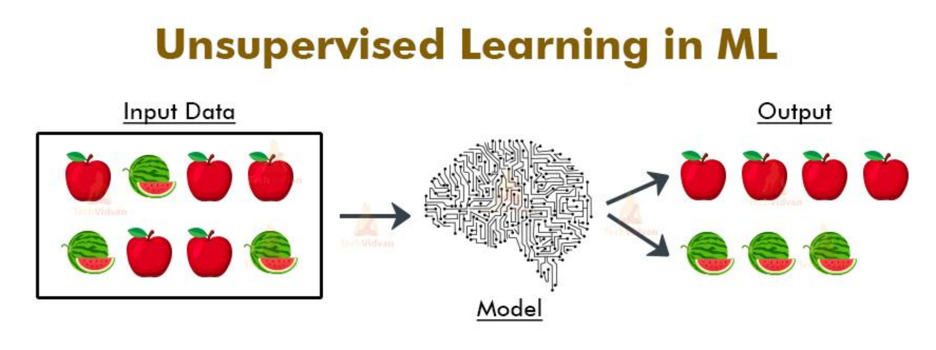




Types of “Learning”

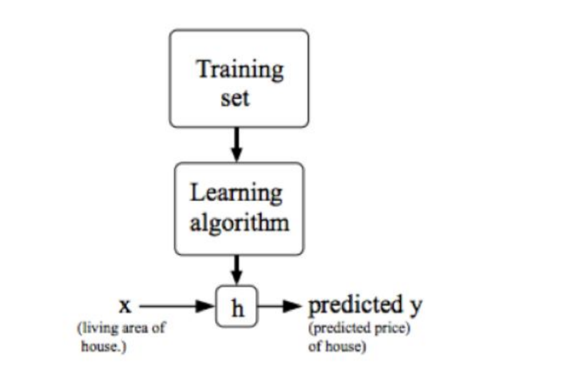
- **Unsupervised Learning**
  - Algorithms trained using **unlabeled** (/unknown) data
  - There is no **feedback**
  - Can be categorized in **Clustering** and **Association**
  - Example: Fruit classifier





Supervised Learning

- Given a training set, learn a function `h(Hypothesis):X → Y` so that `h(x)` is a "good" predictor for the corresponding value of `y`



Regression: When the target variable is continuous - Housing Price Prediction

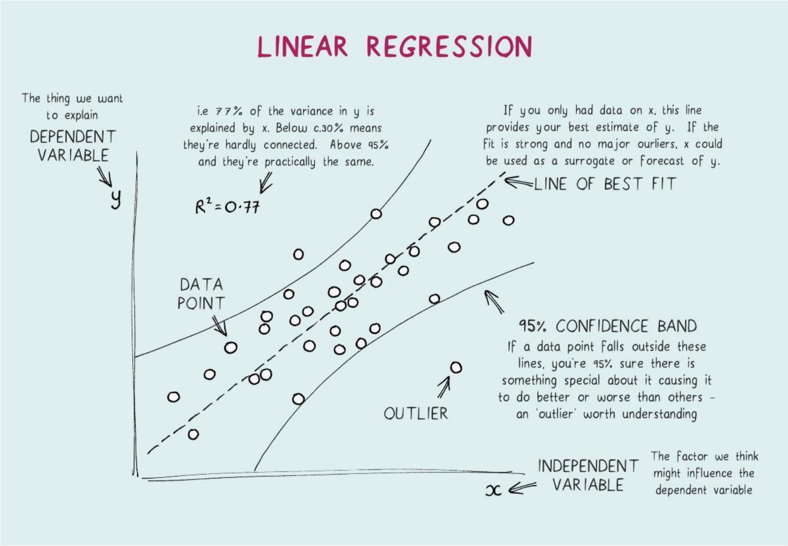
Classification: When the target variable can only take a small number of discrete values such as predicting if it's a house or an apartment

Linear Regression

- Model the relationship between two variables by fitting a **linear equation** to **observed data**
- A **continuous variable** being available
- **Estimating** the dependant

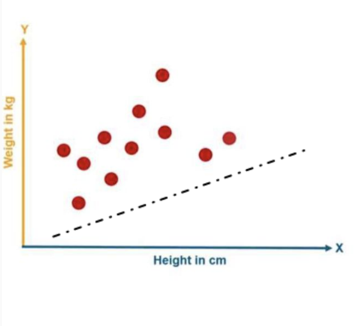
The general equation of a linear regression would be

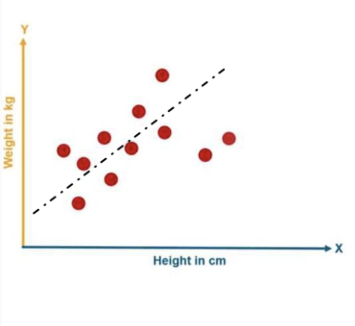
`y = ax + b`

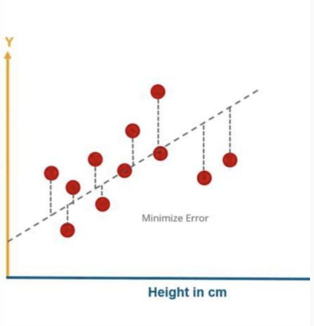


Example

- Start with a random line `w = h`
- Check error in prediction (hypothesis) and update
- Stop when no change or threshold is reached





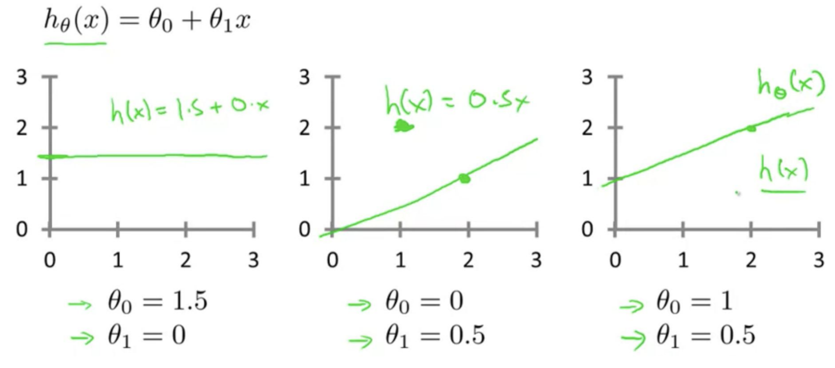


Hypothesis

Training Set

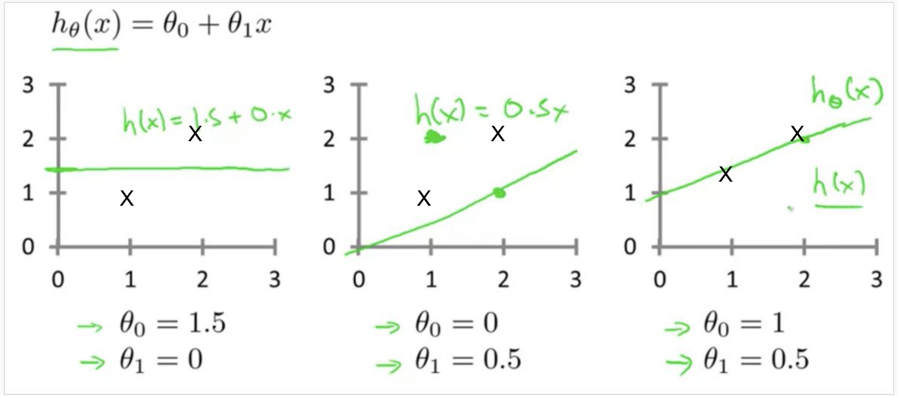
Size in feet² (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$



Cost Function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}_i - y_i)^2 = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x_i) - y_i)^2$$



Cost: $((1.5-1)^2 + (1.5-2)^2)/2*2$
$= 0.5/4$

Cost: ?

Cost: $((1-1)^2 + (2-2)^2)/2*2$
$= 0$

Cost Function

- Accuracy of hypothesis can be measured by using a **cost function**
- In the previous example, select the values such that the cost is minimum

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}_i - y_i)^2 = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x_i) - y_i)^2$$

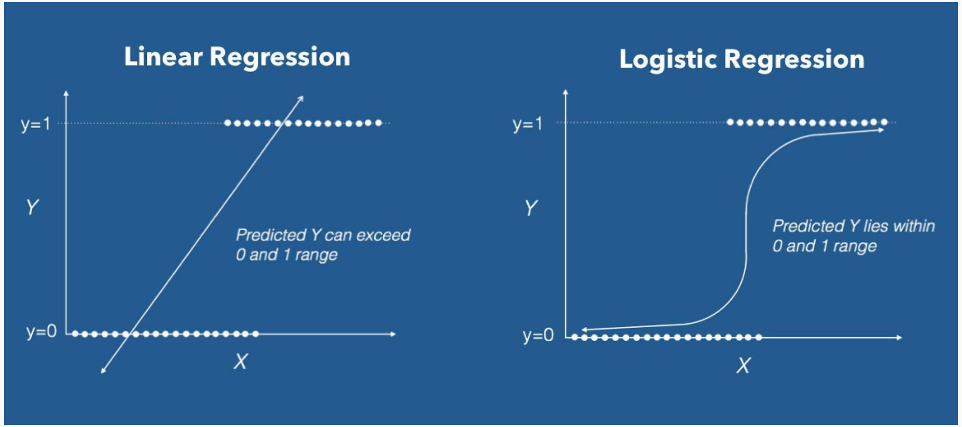
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

- It is an **average of squared difference** between actual output (y) and the hypothesis results in the input (x)

Logistic Regression

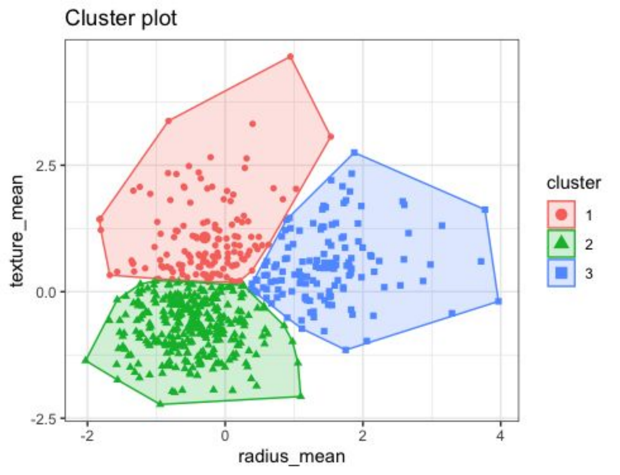
- Modeling the **probability** of a certain class or event
- Can be extended to model **several classes of events**
- Generally used for **classification**
- Has a categorical response

$$P(\text{Event}) = \frac{P(\text{Occurrence})}{P(\text{Not Occurrence})}$$



Clustering

- Automatically discovering **natural grouping** in data
- Unlike **supervised learning**, clustering only interprets the input data and find natural clusters in feature space



Model Representation

$x^{(i)}$	$\Leftarrow$	Input variables / features
$y^{(i)}$	$\Leftarrow$	Output variables / features
$(x^{(i)}, y^{(i)})$	$\Leftarrow$	Training Example
X/x	$\Leftarrow$	Space of input values
Y/y	$\Leftarrow$	Space of output values

Encoding

- Converting categorical data into numerical data to fit and evaluate in the model

- There are multiple methods of encoding

Label Encoding

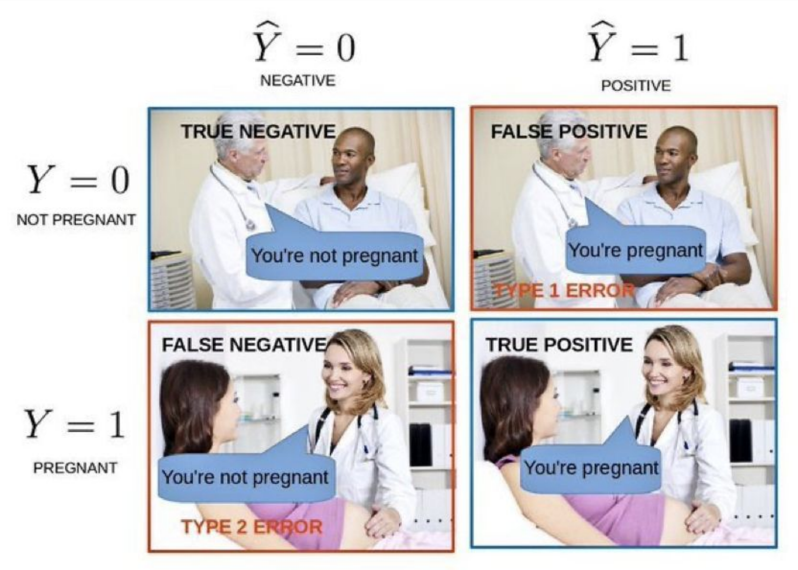
Food Name	Categorical #	Calories
Apple	1	95
Chicken	2	231
Broccoli	3	50

→

One Hot Encoding

Apple	Chicken	Broccoli	Calories
1	0	0	95
0	1	0	231
0	0	1	50

Confusion Matrix



		Predicted class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

Summary

- Machine Learning is a vast topic
- There’s a lot of math!
  - A working knowledge of statistics is required to understand Machine Learning

That’s all Folks!

See you in the next session :)

Give us a feedback: https://bit.ly/3g6ZDlD